

Week 1: Interview Prep

Core Infra • OS • Networking • SSH • Terraform Basics

SECTION 1: OS & SYSTEM BASICS

Q1. What happens when a Linux system boots?

Answer:

- BIOS/UEFI → loads bootloader (GRUB)
- The bootloader loads the Linux kernel
- Kernel initializes hardware + mounts root FS
- systemd starts services

Q2. What is systemd?

Answer:

- Service manager for Linux
- Manages services, logs, and boot sequence

Commands:

- `systemctl start nginx`
- `systemctl status nginx`
- `journalctl -u nginx`

Q3. How do you debug a failed service?

Answer:

1. Check status: `systemctl status <service>`
2. Check logs: `journalctl -u <service>`
3. Check process: `ps aux | grep <service>`

Q4. Difference between top and ps?

- ps → snapshot
- top → real-time

Q5. What is load average?

Number of processes waiting for the CPU

- 1.0 per CPU = healthy
- CPU count = overload

SECTION 2: NETWORKING & DNS (MOST IMPORTANT)

Q1. Difference between ping, curl, and dig?

Q2. How does DNS resolution work?

Flow: Client → Resolver (/etc/resolv.conf) → DNS server → response

Q3. What is /etc/resolv.conf?

Defines DNS servers used by system

Example: nameserver 8.8.8.8

Q4. curl is failing, but ping works; why?

Possible reasons:

- DNS issue
- HTTP service down
- firewall blocking port 80/443

Q5. dig fails; what does it mean?

DNS resolution failure → root cause likely DNS server

SECTION 3: SSH & ACCESS MANAGEMENT

Q1. How does SSH authentication work?

Steps:

1. Client connects
2. Server shares key
3. Client proves identity (private key)
4. Server verifies public key

Q2. What is key-based authentication?

More secure than a password

Uses:

- private key (client)
- public key (server)

Q3. SSH is slow; what could be the issue?

Common causes:

- DNS reverse lookup delay
- high CPU
- PAM delays
- network latency

Q4. How do you debug SSH issues?

`ssh -vvv user@host`

Check:

- connection
- auth
- delays

Q5. What is sudo?

Allows controlled privilege escalation

SECTION 4: TERRAFORM BASICS

Q1. Terraform workflow?

- terraform init
- terraform plan
- terraform apply

Q2. What does Terraform plan do?

Shows what will change (no actual changes)

Q3. Why use Terraform?

Benefits:

- reproducibility
- version control
- automation

Q4. Difference between manual vs IaC infra?

Manual → error-prone

IaC → consistent + scalable



SECTION 5: REAL SCENARIO QUESTIONS

Scenario 1

“ping works but curl fails”

Answer:

- DNS working
- network working
- app/service issue

Scenario 2

“dig fails, everything else fails”

Answer:

- DNS root cause
- fix /etc/resolv.conf

Scenario 3

“SSH stuck after login”

Answer:

- DNS reverse lookup
- PAM
- system load

Scenario 4

“Terraform works locally but not in CI”

Answer:

- missing credentials
- backend config
- environment mismatch





THE DEVOPS WAR ROOM